Gray-Zone Maritime Control in the First Island Chain: A Networked Escalation Game

Workshop 1 Report (Deliverable 2)

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1 Setting and research questions

Contested system. China and the United States compete for day-to-day control of the waters inside the First Island Chain (Senkakus, Taiwan Strait, Luzon Strait, Scarborough and Second Thomas Shoals, Paracels) *below the threshold of war*. China keeps persistent presence with coast guard and maritime militia (CCG) hulls backed by navy (PLAN) ships; the U.S. keeps a smaller number of destroyers (DDG) on presence and freedom-of-navigation patrols (Helmus et al., 2024; Green et al., 2017). Each side decides, every two days, *where* to station its ships on a network of zones and *how* to behave there: patrol (P) or assert (A: shadowing, blocking, water cannon, close maneuvering).

Why strategic interaction is central. Control of a zone depends on the local force ratio, so each side's best allocation depends on the other's; escalation risk is joint (incidents need both sides present, and war costs both); and neither side sees the other's whole fleet.

Research questions.

RQ1. *Control vs. escalation.* Each side's allocation is a MIP with an escalation-risk budget ε_i. How much control does a larger ε_i buy, at what war risk, and which budgets are equilibrium choices when both sides pick them?

RQ2. *Strategic vs. heuristic play.* Do game-theoretic policies (equilibrium and maximin strategies, exact best responses) beat random and rule-based ones, and can mutual restraint be sustained in repeated play?

RQ3. *Robustness and redesign.* Is the best policy or Blue design under nominal conditions also the most robust to weather (Nature's scenario ω), in expected and worst-case terms?

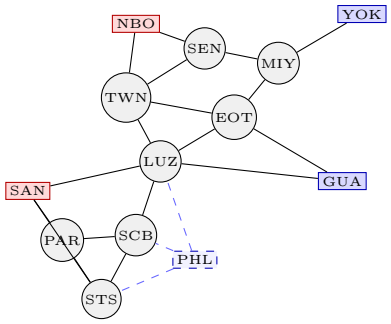
Outcomes (averaged over 200 seeds): Blue's value-weighted control share $\bar{c}_{\mathsf{B}} = \frac{1}{\tau}\sum_t \sum_v V_v^{\mathsf{B}} c_v^t / \sum_v V_v^{\mathsf{B}} \in [-1, 1]$, war rate, payoffs $u_{\mathsf{B}}, u_{\mathsf{R}}$, and for RQ3 the expected and worst-case payoff over ω.

Changes from the proposal and how we used the feedback.

Feedback	What we changed (section)
Model too high level; define it mathematically	Full game on a graph: state, actions, information, dynamics, payoffs (§3)
Q1 is a feasible region; can be a MIP	Best response solved exactly as a MIP; the old aggressive/moderate/conservative types are values of its risk budget ε_i (§3.4)
Research questions more ambitious	RQ1: frontier and equilibrium of a budget game; RQ2: equilibrium play; RQ3: robustness and redesign
Define the resilient redesign	Three Blue designs (§3.5), tested in §4.5
Propose values for missing data	All values in Table 1, marked as assumptions
Show preliminary results	Analytical results and 77 simulated configurations × 200 seeds (§4)
Prof. Sapsis: add external uncertainty (weather) and test robustness	Sea state is Nature's move; four weather scenarios; expected and worst-case payoff (§3.2, §4.5)

2 Literature and current practice

Documented practice. Helmus et al. (2024) catalog China's gray-zone tactics: persistent CCG and militia presence, shadowing, blocking resupply at Second Thomas Shoal, water cannon, and non-navy hulls that stay below the threshold of armed response. Green et al. (2017) argue that deterrence below war depends on presence and on managing escalation. U.S. practice is routine presence and freedom-of-navigation patrols, with encounters managed by the Code for Unplanned Encounters at Sea (WPNS, 2014). We take from these the action set (presence, patrol vs. assert), the asymmetric hull types, and the idea that incidents, not shots, drive escalation. **Models.** Wasser et al. (2019) design a seminar gray-zone game with explicit objectives and action menus; we make every adjudication rule computable. The U.S. Coast Guard schedules randomized patrols from a game model (Shieh et al., 2012); zero-sum network games model transit and blockade (Washburn and Wood, 1995); escalation as a shared, manipulated risk (Schelling, 1966) is what our incident hazard makes explicit. **What we add:** a general-sum game where both sides



One period t (timeline).

1. Nature draws the sea state w_r^t of each region (scenario ω).
2. Each player observes its own ships, ℓ^t, c^t, w^t, and detects opponent ships with probability $q_{i,v}$.
3. Ships whose endurance is nearly spent are forced toward home.
4. Both players choose moves and postures *simultaneously*.
5. Moves resolve under fuel budgets; endurance updates.
6. Control indices update; incidents are drawn; ℓ moves up or down.
7. Stage payoffs are paid; the game stops if $\ell = L$ (war) or $t = T = 60$.

Figure 1: Left: theater graph G (schematic). Squares: home ports (blue H_{B}, red H_{R}); circles: contested zones. Dashed: Subic/EDCA port and edges added by redesign D1. Right: sequence of play.

contest control at once and war is a joint cost, with explicit rules so strategies and designs can be compared over thousands of runs. Numerical values are our assumptions (Table 1), not documented data.

3 Game formulation

3.1 Agents, network and state

Players $\mathcal{I} = \{\mathsf{B}, \mathsf{R}\}$. Ship types $\mathcal{K}_{\mathsf{B}} = \{\text{DDG}\}$ (10 hulls), $\mathcal{K}_{\mathsf{R}} = \{\text{PLAN}, \text{CCG}\}$ (8 and 6). The theater is an undirected graph $G = (\mathcal{V}, \mathcal{E})$ (Fig. 1) with Blue home ports $H_{\mathsf{B}} = \{\text{YOK}, \text{GUA}\}$, Red home ports $H_{\mathsf{R}} = \{\text{NBO}, \text{SAN}\}$ and eight contested zones $\mathcal{C}$ grouped into regions North $= \{\text{SEN}, \text{MIY}\}$, Center $= \{\text{TWN}, \text{EOT}, \text{LUZ}\}$, South $= \{\text{SCB}, \text{STS}, \text{PAR}\}$. One period is two days; an edge takes one period. The state at period t is $s^t = (x^t, d^t, c^t, \ell^t, w^t)$, where $x_{k,v}^t \in \mathbb{Z}_{\geq 0}$ counts type-k ships at node v; d^t gives each ship's remaining endurance (periods at sea, at most D_k); $c_v^t \in [-1, 1]$ is the control index of zone v (+1 Blue, −1 Red); $\ell^t \in \{0, \ldots, L\}$ is the escalation level ($L = 5$ is war); and $w_r^t \in \{\text{calm}, \text{rough}, \text{storm}\}$ is the sea state of region r.

3.2 Actions, constraints, information and uncertainty

Actions. Each period player i chooses for every free ship a destination at most one edge away (staying is allowed) and a posture $p_{i,v} \in \{\text{P}, \text{A}\}$ at every zone it occupies. Writing $f_{k,uv}^t$ for the number of type-k ships moving from u to v, the feasible set is defined by

$$
\begin{aligned}
& x_{k,u}^{t+1} = x_{k,u}^t + \textstyle\sum_{v \in \delta(u)} f_{k,vu}^t - \sum_{v \in \delta(u)} f_{k,uv}^t && \text{(flow conservation, Lecture 2)} \\
& \textstyle\sum_{k,(u,v)} f_{k,uv}^t \leq F_i && \text{(fuel: } F_{\mathsf{B}} = 8,\ F_{\mathsf{R}} = 10 \text{ moves/period)} \\
& d \leq \text{dist}(v, H_i) \Rightarrow \text{ship heads home};\ d \leftarrow D_k \text{ after a period in port} && \text{(endurance)}
\end{aligned}
$$

and no ship may enter the opponent's home port or a storm-closed zone.

Information. Each player observes its own ships, c^t, ℓ^t and w^t (public). It detects the opponent's ships at v with probability $q_{i,v} = 0.95$ if it has a ship at v, 0.6 if adjacent, 0.2 otherwise (satellite); Red has $q \geq 0.6$ at four zones near its coast; rough or storm seas multiply q by 0.7. A player's *belief* is the last detected count at each node, so players move simultaneously at information sets that do not reveal the opponent's positions (imperfect monitoring, Lectures 5–8).

Uncertainty (Nature). Weather: in scenario $\omega \in \Omega = \{\text{calm}, \text{nominal}, \text{typhoon}, \text{extreme}\}$ (weights $\pi_\omega = (0.25, 0.40, 0.25, 0.10)$, Lecture 1), each region starts a 3-period storm w.p. $p_s(\omega) \in \{0, 0.02, 0.06, 0.12\}$ and is otherwise rough w.p. $p_r(\omega) \in \{0, 0.2, 0.35, 0.5\}$. A storm closes the region's zones and freezes their control; rough seas halve CCG effectiveness and degrade detection. Incidents are also random (below).

3.3 Transition dynamics, payoffs and termination

Control. Local strength is $S_{i,v} = \sum_{k \in \mathcal{K}_i} w_k \, \alpha^{\mathbf{1}[p_{i,v} = \text{A}]} x_{k,v}$ with $w_{\text{DDG}} = w_{\text{PLAN}} = 1$, $w_{\text{CCG}} = 0.6$ (0.3 in rough seas), $\alpha = 1.5$. With $\phi_v = (S_{\mathsf{B},v} - S_{\mathsf{R},v})/(S_{\mathsf{B},v} + S_{\mathsf{R},v} + \kappa)$,

$$
c_v^{t+1} = (1 - \lambda) c_v^t + \lambda \phi_v, \qquad \lambda = 0.3,\ \kappa = 1, \tag{1}
$$

so control is earned by persistent presence and decays when a side leaves.

Incidents and escalation. Where both sides are present, an incident occurs with probability $\rho_v = 1 - e^{-h_v}$, where

$$
h_v = \left(h_0 + h_{\text{A}}^{\mathsf{B}} \mathbf{1}[p_{\mathsf{B},v} = \text{A}] + h_{\text{A}}^{\mathsf{R}} \mathbf{1}[p_{\mathsf{R},v} = \text{A}]\right) \min\{S_{\mathsf{B},v}, S_{\mathsf{R},v}\} \, (1 + \ell^t/2). \tag{2}
$$

Zone	TWN	SEN	MIY	EOT	LUZ	SCB	STS	PAR	Other parameters
V_v^{B}	10	6	4	4	6	5	5	2	$T=60$, $\delta=0.98$, $\gamma=0.1$, $\eta=1$, $W=200$, $\ell^1=1$
V_v^{R}	10	6	3	4	5	7	6	6	$h_0=0.006$, $h_{\mathrm{A}}^{\mathsf{B}}=0.03$, $h_{\mathrm{A}}^{\mathsf{R}}=0.036$ (PLAN), 0.012 (CCG)
c_v^1	0	0.2	0.3	0.2	0.1	−0.2	0	−0.4	$D_{\mathrm{DDG}}=10$, $D_{\mathrm{PLAN}}=8$, $D_{\mathrm{CCG}}=6$, $\Lambda=20$

Table 1: Proposed parameter values (assumptions; no public data). Hazards and μ were set so that war is uncommon but not rare (0–34% of games across the configurations tested).

Each incident raises ℓ by one; a period without incidents lowers ℓ by one with probability $\mu = 0.25$. The factor $(1+\ell/2)$ makes crises self-reinforcing.

Payoffs. With $\sigma_{\mathsf{B}}=+1$, $\sigma_{\mathsf{R}}=-1$ and m_i^t the number of moves,

$$g_i^t = \sigma_i \textstyle\sum_{v\in\mathcal{C}} V_v^i\, c_v^{t+1} - \gamma\, m_i^t - \eta\, \ell^{t+1}, \qquad u_i = \textstyle\sum_{t=1}^{\tau} \delta^{t-1} g_i^t - W\,\mathbf{1}[\ell^\tau = L], \tag{3}$$

where $\tau = \min\{T,\ \text{first } t \text{ with } \ell^t = L\}$ (termination by war or horizon). The war penalty W hits both players, which makes the game general-sum. Zone values V_v^i differ by player (Table 1).

3.4 Decision models: the best-response MIP and the template game

Best response as a MIP (Lecture 2). Given its beliefs $\hat{x}_{-i}$ and the sea state, player i chooses binary moves $y_{s,w}$ (ship s goes to node w) and, at every zone, one binary option $b_{v,(n,p)}$ meaning "I will have n hulls at v with posture p". Because ϕ_v and ρ_v are nonlinear in the counts, we tabulate them for every option, which makes the problem linear:

$$\begin{aligned}
\max_{y,b,x}\ & \textstyle\sum_{v\in\mathcal{C}} \sum_{(n,p)} a_v(n,p)\, b_{v,(n,p)} - \gamma \sum_{s,\, w\neq \mathrm{loc}(s)} y_{s,w} \\
\text{s.t.}\ & \textstyle\sum_w y_{s,w} = 1 \ \ \forall s; \quad \sum_{s,\, w\neq\mathrm{loc}(s)} y_{s,w} \le F_i; \quad x_{k,v} = \bar{x}_{k,v} + \sum_{s \text{ of type } k} y_{s,v}; \\
& \textstyle\sum_{(n,p)} b_{v,(n,p)} = 1, \quad \sum_{(n,p)} n_k\, b_{v,(n,p)} = x_{k,v} \ \ \forall v,k; \qquad \sum_v \sum_{(n,p)} \rho_v(n,p)\, b_{v,(n,p)} \le \varepsilon_i\,(1-\ell^t/L); \\
& y, b \in \{0,1\},
\end{aligned}$$

where $a_v(n,p) = V_v^i \lambda H\, \sigma_i \phi_v(n,p) - \Lambda \rho_v(n,p)$, $H = 1/(1-\delta(1-\lambda))$ converts a one-period change in ϕ into its discounted effect on control through (1), $\Lambda = 20$ is the cost a player assigns to one expected incident, and $\bar{x}$ counts ships on forced returns. The last constraint is the **escalation-risk budget**: its feasible region is exactly the set of allocations whose expected number of incidents stays below ε_i, tightened as ℓ rises. The proposal's "conservative, moderate, aggressive" types become $\varepsilon_i \in \{0.05, 0.15, 0.40\}$. We solve it with Gurobi, as in the lecture code.

Template game (Lectures 3–4). To compute equilibria we abstract each player's strategy to one of 14 templates: a split of deployable ships across North/Center/South (seven splits) times a posture (P or A). This gives a finite strategic-form game $\Gamma = \langle \mathcal{I}, (S_i), (u_i)\rangle$ with $|S_i| = 14$ whose payoffs $M_i(a_{\mathsf{B}}, a_{\mathsf{R}}) = \sigma_i \sum_v V_v^i \phi_v - \Lambda \sum_v \rho_v$ are evaluated with (1)–(2) at steady state (calm seas, $\ell = 1$). We analyze it with iterated elimination of strictly dominated strategies (IESDS), pure-NE enumeration, the maximin LP, and the correlated-equilibrium LP.

Policies. *random*; *moderate* (rule-based: asserts only with local superiority); *greedy* (fast MIP heuristic); *MIP-BR*(ε); *equilibrium* (its part of a pure NE of Γ); *maximin* (samples its maximin mixed strategy of Γ).

3.5 Resilient redesigns (Blue)

Each changes the system, not the operating policy: **D1 dispersed basing** adds a Blue port at Subic (EDCA access) with edges to SCB, LUZ, STS (alternative paths and shorter cycles; cost 0.3/period); **D2 redundant sensing** gives Blue a detection floor of 0.6 in every zone, also in bad weather (cost 0.5/period); **D3 crisis hotline** (CUES-style communication) raises the de-escalation probability to $\mu = 0.5$ and halves assert hazards once $\ell \ge 3$ (no cost to Blue, but it also helps Red).

Simplifying assumptions. Two unitary players; control is a force-ratio index; incidents are the only path to war; beliefs are last-seen counts; best responses look one period ahead.

4 Preliminary results

§4.1–4.2 are analytical (template game); the rest use 200 seeds per configuration, the same for every policy. Unless stated, Red plays MIP-BR with $\varepsilon_{\mathsf{R}} = 0.15$ in nominal weather.

4.1 Small example: a game of chicken, and sustainable restraint

Fix the force splits that survive dominance (§4.2): Blue center-heavy, Red south-heavy. Each side then only chooses P or A. The resulting 2×2 sub-matrix of Γ, normalized so (P,P) = (0, 0), has (P,A) = (−9.61, 1.20), (A,P)

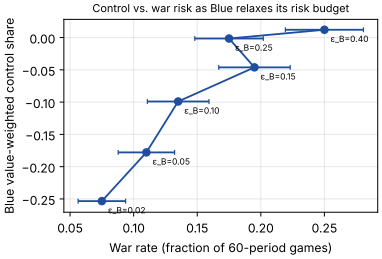
= (0.28, −10.45) and (A,A) = (−11.41, −11.33) (Blue, Red). Asserting against a patroller pays a little ($G_{\mathsf{B}} = 0.28$, $G_{\mathsf{R}} = 1.20$), but mutual assertion costs more than patrolling alone, so $BR_i(\mathrm{P}) = \mathrm{A}$ and $BR_i(\mathrm{A}) = \mathrm{P}$: the anti-coordination (chicken) structure of the shared-compute example (Lecture 3). There are two pure NE, (A,P) and (P,A), and a mixed NE (Blue asserts w.p. 0.58, Red 0.13) with mutual assertion w.p. 0.08. The welfare-maximizing correlated equilibrium (CE LP) puts 0.40 on (P,P), 0.54 on (A,P), 0.06 on (P,A) and *zero* on (A,A): a mediator (e.g. a hotline) removes mutual assertion.

In the repeated game (Lectures 5–6), consider grim trigger: both patrol; after any assertion, both play the mixed NE forever. By the one-stage deviation principle, patrolling is optimal iff $(1-\delta)G_i + \delta v_i \leq 0$, i.e. $\delta \geq G_i/(G_i - v_i)$, which is 0.18 for Blue and 0.17 for Red. With $\delta = 0.98$, **mutual restraint is a subgame perfect equilibrium**, although our one-period-ahead policies cannot see this yet.

4.2 Template game: dominance, equilibria and the effect of the escalation cost

IESDS reduces Γ from 14×14 to 2×4: Blue keeps only the center-heavy split (N20/C60/S20), Red the south- or center/south-heavy splits. There are two pure NE: (Blue C-heavy assert, Red S-heavy patrol), payoffs $(0.39, -9.27)$, and (Blue C-heavy patrol, Red C/S assert), $(-14.44, 2.71)$; the *equilibrium* policy plays the first (higher total payoff; this selection is an assumption). Without the escalation cost, the zero-sum maximin LP has a pure saddle point where *both* sides concentrate on the Center and assert (value −2.26): the shared incident cost is what makes at most one side assert.

4.3 RQ1: control versus war risk, and the risk-budget game



B\R	cons. 0.05	mod. 0.15	aggr. 0.40
cons. 0.05	−194, 199 war 0.10	−303, 315 0.11	−308, 318 0.12
mod. 0.15	18, −36 war 0.15	−144, 101 0.20	−192, 140 0.21
aggr. 0.40	74, −80 war 0.14	−83, 16 0.25	**−151, 39** **0.34**

Figure 2: Left: Blue's control share vs. war rate as Blue's risk budget ε_{B} grows (Red fixed at 0.15). Right: the risk-budget game, $(u_{\mathsf{B}}, u_{\mathsf{R}})$ and war rate when both sides choose ε (s.e. $\approx$ 5–12). Bold: the unique Nash equilibrium.

Frontier. Raising ε_{B} from 0.02 to 0.15 lifts Blue's control share from −0.25 to −0.05 while the war rate rises from 0.07 to 0.20; going to 0.40 adds only 0.06 of control (Fig. 2). Most of what a looser budget buys is bought by $\varepsilon \approx 0.15$–0.25.

Risk-budget game. When both sides pick a budget, "aggressive" is a best response to every opponent choice for both players (Red vs. conservative Blue, 318 vs. 315, is within noise), so (aggressive, aggressive) is the unique pure NE. It has the *highest* war rate (0.34 vs. 0.10 when both are conservative) and total payoff −112 vs. +12 at (conservative, moderate): a prisoner's dilemma (Lecture 3). So restraint must come from repetition or coordination (§4.1).

4.4 RQ2: strategic versus heuristic play

Blue \ Red	rule-based (moderate)	MIP-BR	equilibrium	worst case over Red
random	−283 / −0.14 / 0.15	−319 / −0.16 / 0.33	−164 / −0.08 / 0.06	−319
rule-based (moderate)	−122 / −0.05 / 0.04	−184 / −0.09 / 0.06	35 / +0.06 / 0.04	−184
greedy	−70 / +0.01 / 0.12	−137 / −0.05 / 0.14	65 / +0.08 / 0.03	−137
MIP-BR	−75 / +0.01 / 0.13	−144 / −0.05 / 0.20	53 / +0.08 / 0.08	−144
equilibrium	−72 / −0.01 / 0.07	**−116** / −0.03 / 0.10	21 / +0.05 / 0.06	**−116**
maximin	−170 / −0.09 / 0.04	−221 / −0.12 / 0.08	−35 / −0.01 / 0.00	−221

Table 2: Blue payoff u_{B} / control share / war rate (s.e. of u_{B} is 1–7). Nominal weather, design D0.

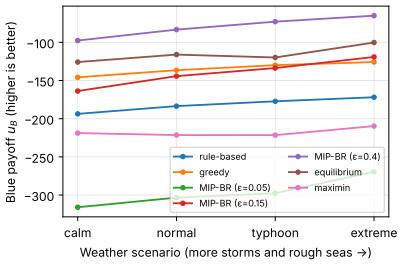
Greedy, MIP-BR and equilibrium beat the random and rule-based benchmarks (Table 2). The **equilibrium** policy has the best worst case over Red's policies (−116) and, against MIP-BR, half the war rate of MIP-BR itself (0.10 vs. 0.20). Greedy and MIP-BR do best against weaker opponents but escalate more. **Maximin** is the safest (war rate ≤ 0.08) but gives up the most control: it guards against a worst-case Red template that Red rarely plays.

MIP vs. greedy. Over 1,524 Blue decisions in 30 games the greedy heuristic matched the MIP optimum in 67%

of decisions but broke the risk budget in 14%, so the MIP is needed to enforce the feasible region exactly; in play the two perform about the same.

4.5 RQ3: robustness to weather and resilient designs

As in Lecture 1, we report each Blue policy or design x's expected payoff $\sum_\omega \pi_\omega J(x,\omega)$ and worst case $\min_\omega J(x,\omega)$.



Design	calm	nom.	typh.	extr.	exp.	worst
D0 baseline	−164/.24	−144/.20	−134/.12	−119/.10	−144	−164
D1 dispersed	−153/.33	−141/.28	−131/.24	**−98**/.14	−137	−153
D2 sensing	−181/.23	−164/.23	−149/.12	−136/.11	−162	−181
D3 hotline	**−140**/.10	**−118**/.05	**−111**/.02	−102/.01	**−120**	**−140**

Figure 3: Left: Blue payoff by policy and weather. Right: designs (MIP-BR vs. MIP-BR), u_B / war rate (s.e. ≈ 7).

Weather. Bad weather helps Blue slightly: most policies do worst in *calm* seas (Fig. 3), since rough seas halve the effect of Red's coast-guard hulls and storms pause operations and incidents (MIP-BR war rate 0.24 calm, 0.10 extreme). The ranking is robust: MIP-BR ($\varepsilon = 0.40$) and equilibrium are first and second in all four scenarios, and the only change below them is a swap within noise. So the best policy on average is also best in the worst case. Caution: the top scorer also has the most wars (0.12–0.29), so $W = 200$ may be too small.

Designs. The hotline (D3) is best in expected and worst-case terms and cuts the war rate fourfold (0.20 → 0.05, nominal). It also raises Red's payoff (101 → 144): fewer wars help both sides. We charged it nothing, so part of this win is by assumption; Workshop 2 will give it a cost. Dispersed basing (D1) barely changes Blue's payoff but cuts Red's (101 → 16), at a higher war rate (0.28). Redundant sensing (D2) shows no measurable benefit even at zero cost: better Blue information barely changes Blue's decisions in this model.

Limitations. Parameters are assumptions; policies look one period ahead; the war rate depends strongly on h, μ and W; payoff differences below about 20 are ties (s.e. 5–12).

5 Prototype

Current state. `wargame_gui.py` runs the model of §3 as a **GUI** (Fig. 4: pick each side's plan, Blue's design and the weather, then step through turns or run many games; a Help button explains the map), as **batch** runs, or as a turn-by-turn **trace**. `experiments.py all` regenerates every number in §4; instructions are in `README.md`.

How the code reflects the model. `Game.step` is the timeline of Fig. 1 with (1)–(3); `MIPBRPolicy` solves the MIP of §3.4 each period; `MetaGame` builds Γ and runs IESDS, NE enumeration and both LPs.

Still to build. Learning agents, an attacker that reacts to Blue, Bayesian beliefs and multi-period policies (§6).

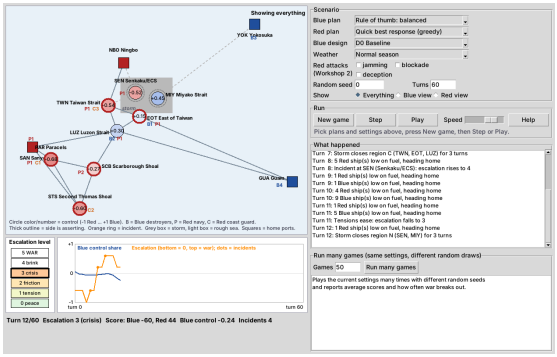


Figure 4: The GUI after 12 turns (storm in the North).

6 Plan for Workshop 2

Model. Bayesian beliefs (Lectures 7–8), multi-period best response, ±50% sweep on V, h, W, μ.

Learning (Module 2, once taught). A Red agent that learns from repeated play (reinforcement learning), and fictitious-play or no-regret updates on the template game. Do they converge or cycle, drift toward restraint or assertion, and exploit the predictable rule-based Blue policy?

Attacks and defenses (Module 3). Run the three coded Red attacks (CCG blockade of Blue's edges at SCB/STS, jamming of Blue sensing, false contacts) and an attacker that moves after seeing Blue's posture (backward induction). Defenses: randomized patrols, D1, D2; metric: change in control and war rate.

Resilient designs (Module 4). D0–D3 and combinations under attack and a rare long typhoon; price the hotline. **Timeline:** Nov 3 beliefs, attacks; Nov 17 learning; Dec 1 redesigns; Dec 8 report.

References

Green, M., Hicks, K., Cooper, Z., Schaus, J., and Douglas, J. (2017). *Countering Coercion in Maritime Asia: The Theory and Practice of Gray Zone Deterrence*. Center for Strategic and International Studies.

Helmus, T. C., Romita Grocholski, K., Liggett, T., Rhoades, A. L., Savitz, S., and Palmer, K. (2024). *Understanding and Countering China's Maritime Gray Zone Operations* (RR-A2954-1). RAND Corporation. https://doi.org/10.7249/RRA2954-1

Schelling, T. C. (1966). *Arms and Influence*. Yale University Press.

Shieh, E., An, B., Yang, R., Tambe, M., Baldwin, C., DiRenzo, J., Maule, B., and Meyer, G. (2012). PROTECT: A deployed game theoretic system to protect the ports of the United States. In *Proc. AAMAS 2012*, 13–20.

Washburn, A., and Wood, K. (1995). Two-person zero-sum games for network interdiction. *Operations Research*, 43(2), 243–251.

Wasser, B., Oberholtzer, J., Pettyjohn, S. L., and Mackenzie, W. (2019). *Gaming Gray Zone Tactics: Design Considerations for a Structured Strategic Game* (RR-2915-A). RAND Corporation. https://doi.org/10.7249/RR2915

Western Pacific Naval Symposium (2014). *Code for Unplanned Encounters at Sea (CUES)*.